Andrew Thomas

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Personal Statement

Electronic and Embedded Engineer at TouchNetix. Skilled in electronics, software development and embedded systems. Holds an MEng in Electrical and Electronic Engineering from Lancaster University and a recipient of the UKESF Scholarship.

Seeking to transition into a long-term, meaningful career aligned with a personal desire to work towards a sustainable future. A quick learner with a broad technical skillset, always striving to make a difference.

Technical Skills

- **Programming Languages**: C, C++, Python (PyTorch, NumPy, Pandas)
- **Embedded Systems**: I2C, SPI, CAN protocols; STM32, Arduino microcontrollers
- **Software Development**: Linux systems, thread management, CMake, Jenkins, Bash
- **Circuit Design**: PCB analogue/digital design (Altium, KiCad); PCB milling and fabrication
- **Simulation Tools**: MATLAB, SolidWorks, Ansys, COMSOL for thermal-mechanical simulation
- **Prototyping & Testing**: Wire-bonding, soldering, laser milling; TFT probing and analysis

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Experience

TouchNetix, Fareham – Embedded Engineer

September 2025 – Present

- Developing a touchscreen controller driver for mainline Linux Kernel (LKML). This involves navigating the strict upstreaming process, adhering to kernel coding standards, and managing patch submissions.
- Collaborating directly with the hardware team during chip bring-up to develop self-test routines. Identified and leveraged underutilized analog front end resources, cutting latency from 2s to 50ms and improving factory throughput.
- Investigated and resolved firmware bugs and I2C/SPI timing issues using logic analysers and oscilloscopes to ensure robust communication between the controller and peripherals.

Accelercomm, Southampton – 5G Software Engineer

October 2023 – September 2025

- Leading company-wide system compliance to the product specification and 3GPP 5G NR standards.
- Developed a new testing framework to validate 3GPP 5G physical channels, generating 5G NR slot/PDU scenarios using MATLAB.
- Implemented DPDK-based hardware accelerator drivers, enhancing system efficiency.
- Acquired in-depth knowledge of standards including 3GPP, O-RAN, nFAPI, and eCPRI.
- Collaborating on code-reviews to ensure code quality standards and consistent approach across the team.

Lancaster Formula Student, Lancaster University – Instrumentation Lead

September 2022 – July 2023

- Designed and tested analogue processing PCBs for the control of the EV power system.

- Led the development of a CAN-based telemetry network, integrating sensors using I2C/SPI with EV systems for real-time data collection.
- Designed and tested PCBs fitted with STM32 microcontrollers configured with RTOS thread management, optimized for multi-priority task handling.

CSA Catapult, Newport – Advanced Packaging Intern

July 2021 – July 2022

- Improved wire bonding and die attachment processes for micro-scale MMIC and VCSEL dies.
- Conducted thermo-mechanical simulations, reducing thermal stress in semiconductor designs.
- Collaborated with industry partners to prototype innovative RF and power semiconductor applications.

Additional Roles

- **Training Captain, Lancaster University Athletics Club** (2020–2021): Designed tailored training plans for diverse skill levels.
- **Volunteer, Morecambe Foodbank & Refugee Community Kitchen** (2019–2020): Coordinated logistics and service delivery under time-sensitive conditions.

Education

Lancaster University - MEng Electrical and Electronic Engineering
Sep 2018 – July 2023
2 Scholarships – overall 2:1

Part 1 – 69.2 %
Modules include: Engineering Management, Electrical Circuits and Power Systems, Electromagnetics & RF Engineering, Digital Electronics, Power Electronics and Applications, Integrated Circuit Engineering, Optoelectronics and Wireless Communications & Digital Signal Processing.

Part 2 – 70.4 %

Modules include: Advanced Embedded Systems, Terahertz Science and Technology, Microengineering & High Frequency Electronics.

Epsom College
September 2012 – July 2018
A-Levels: Maths (A*), Further Maths (A*), Physics (A*), Chemistry (A)
GCSEs: 10 subjects including Maths, Physics, and Chemistry (A*/A/B)

Projects & Interests

- **AI Development**: Built convolutional neural networks for image classification using Kaggle datasets.
- **Hobbyist Projects**: Explored microcontroller communication protocols and designed small-scale PCB systems.
- **Scientific Research Enthusiast**: Regularly follows advancements in physics, cosmology, and AI via sources such as Sabine Hossenfelder and PBS Space Time.
- **Personal Pursuits:** A keen runner, ranked in the Top 1000 in the UK (Half Marathon 1hr10, 5k 15:30). Self-taught classical piano player.